KYLE DAWSON, PHD

Lead Data Scientist | Optimization, Forecasting & Production ML for the Energy Sector

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new geospatial intelligence (GEOINT) team, setting technical direction, mathematical-modeling and evaluation standards, and mentoring scientists and engineers across machine-learning and optimization initiatives
- directed the development and rollout of a large-scale budget-constrained optimization model (mixed-integer linear programming via CBC with greedy heuristics, LightGBM outage-risk scoring, LiDAR features on BigQuery), in partnership with external consultants, that allocates a fixed tree-trimming budget to maximize grid reliability (SAIFI) per dollar
- built and deployed production-grade Python systems on GCP (Cloud Run, Docker, CI/CD) that translate ambiguous grid-reliability and asset-management problems into optimization and forecasting models used by operations and business stakeholders

Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- designed, deployed, and maintained customer-facing analytics platforms (Plotly Dash on Azure/Android) delivering real-time greenhouse-gas quantification, QA/QC, and decision support, securing $50-150k contracts
- built and productionized machine-learning models to forecast survey quality, detect anomalies, and constrain uncertainty in methane and CO_2 emission estimates
- implemented Bayesian uncertainty propagation and physics-based boundary-layer simulation to generate decision-grade emissions metrics used in regulatory and enterprise reporting
- authored technical documentation, uncertainty assessments, and customer-facing deliverables supporting multi-year enterprise contracts across 10+ countries
- presented field-deployed modeling systems and results to executive leadership, customers, and international conference audiences

Data Scientist — Ag Tech

Bayer CropScience

Feb 2020 – Mar 2022 St. Louis, MO

- applied machine learning and geospatial analytics to UAV-derived imagery for large-scale digital phenotyping and crop performance assessment
- built end-to-end analytics workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets, with interactive Plotly Dash apps delivering decision support to stakeholders

Postdoctoral Fellow

NASA Langley Research Center

Oct 2017 – Feb 2020 Hampton, VA

- developed uncertainty-aware aerosol retrieval models from HSRL observations and delivered peer-reviewed results for climate applications

SELECTED PROJECT

BESS Day-Ahead Arbitrage Optimizer

Self-directed proof of concept

2025

EXECUTIVE SUMMARY

Data science leader and PhD physical scientist with 8+ years building and productionizing machine-learning, forecasting, and physics-based simulation systems on heterogeneous sensor, spatial, and time-series data, and turning them into scalable, production-grade Python software in the cloud. Increasingly focused on mathematical optimization for the energy sector — directing a large-scale budget-constrained MILP for grid reliability and prototyping wholesale-market (ERCOT) optimization. Skilled at partnering with product, engineering, and commercial stakeholders to translate complex problems into rigorous models and shipped software, while setting technical direction and mentoring scientists and engineers.

CORE COMPETENCIES

Technical Leadership · Mentoring & Team Building · Mathematical Modeling · Research-to-Production · Product & Customer Translation · Model Evaluation & Validation · Cross-Functional Collaboration · Executive & Stakeholder Communication · Technical Documentation

DOMAIN EXPERTISE

Mathematical Optimization · Mixed-Integer Linear Programming · Constrained & Combinatorial Optimization · Operations Research · Forecasting & Time Series · Simulation & Physics-Based Modeling · Machine Learning · Anomaly Detection · Bayesian Inference & Uncertainty Quantification · Power & Energy Systems · Geospatial Analytics · Remote Sensing

TOOLS & TECHNOLOGIES

Python · MILP / Optimization Solvers · SciPy · NumPy · Pandas · Scikit-Learn · LightGBM · XGBoost · PyTorch · SQL · GCP BigQuery · Cloud Run · Vertex AI · AWS · Azure · Docker · REST APIs · MLOps / CI-CD · Git · Plotly

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 Hampton, VA

A PuLP/CBC linear program, fed by ERCOT API data, that schedules day-ahead battery charge/discharge arbitrage against nodal settlement-point (LMP) prices under state-of-charge and power constraints.

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 | Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing